

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Q. No	Understanding of Concepts (25)	Experimental Design (30)	Use of Tools (20)	Analysis of Results (15)	Documentation (10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

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Application-Based Questions

1) student organization wants to send the message "MEET AT NOON" using Caesar cipher with a shift of 4.

a) Encryption Process.

$$M(12) + 4 = 16 \rightarrow Q$$

$$E(4) + 4 = 8 \rightarrow I$$

$$E(4) + 4 = 8 \rightarrow I$$

$$T(19) + 4 = 23 \rightarrow X$$

$$A(0) + 4 = 4 \rightarrow E$$

$$T(19) + 4 = 23 \rightarrow X$$

$$N(13) + 4 = 17 \rightarrow R$$

$$O(14) + 4 = 18 \rightarrow S$$

$$O(14) + 4 = 18 \rightarrow S$$

$$N(13) + 4 = 17 \rightarrow R$$

cipher text = Q I I X E X R S S R.

b) Decryption:

$$Q(16) - 4 = 12 \rightarrow M$$

$$I(8) - 4 = 4 \rightarrow E$$

$$I(8) - 4 = 4 \rightarrow E$$

$$X(23) - 4 = 19 \rightarrow T$$

$$E(4) - 4 = 0 \rightarrow A$$

$X(23) - 4 = 19 \rightarrow T$
 $P(17) - 4 = 13 \rightarrow N$
 $S(18) - 4 = 14 \rightarrow O$
 $S(18) - 4 = 14 \rightarrow O$
 $R(17) - 4 = 13 \rightarrow N$

Q11E EX ESSE $\rightarrow$ MEET AT NOON

3) Plaintext = HELLO
Keyword = KEY

A company encrypted the word "Hello" using the Vigenere cipher with the keyword "key".

a) Encryption Process.

Repeat the key word until key word matches the length of plaintext.

plaintext = H E L L O

key word = K E Y K E

H E L L O
7 4 11 11 14

K E Y K E

10 4 24 10 4

Add.

$$7 + 10 = 17 \Rightarrow R$$

$$4 + 4 = 8 \Rightarrow I$$

$$11 + 24 = 35 \Rightarrow J \quad 26 + 7 = 35$$

$$11 + 10 = 21 \Rightarrow V$$

$$4 + 4 = 8 \Rightarrow I$$

HELLO = RIJVS

Cipher text is RIJVS

b) Because the shift changes for every letter, the encryption is stronger than a Caesar cipher.

5) A legacy communication system with transposition cipher (3 level arrangement)

Cipher text: WECLTEERDSOI
AOLAIJVE

W	E	C	R	L	T	F	E	R
D	S	O	F	I	F	E	A	O
C	A	J	V	D	F	N	C	E

2) GSRN RH Z HVXIVG

a) Deduce the most likely original message

albash cipher

A	- Z	K	- P	U	- F
B	- Y	L	- O	V	- E
C	- X	M	- N	W	- D
D	- W	N	- M	X	- C
E	- V	O	- L	Y	- B
F	- U	P	- K	Z	- A
G	- T	Q	- J		
H	- S	R	- I		
I	- R	S	- H		
J	- Q	T	- G		

G - T R - I
 S - H H - S Z - A
 R - I
 L - S

H - G
 V - E
 X - C
 I - R
 V - E
 G - T

Original message

GSRH RH Z HVXIVG

Decryption.

This is a secret.

result:

THIS IS A SECRET

b) $\Rightarrow$ RH appears repeatedly and decrypts to IS, a common English word

$\Rightarrow$ Z decrypts to A, a common one-letter word.

e) The word lengths match normal English words.

f) The atbash substitution ($A \rightarrow Z$, $B \rightarrow Y$, $C \rightarrow X, \dots$) correctly converts the entire message.

This is a secret.

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